

Loan Charge-Off-Analysis

Deceased Member Impact

Tools used SQL | Power BI

Executive Summary

This project analyzes loan charge-offs within a credit union, focusing on accounts linked to deceased members to assess financial impact and identify key risk drivers. Using SQL and Power BI, the analysis reveals patterns in loan types, insurance coverage, and recovery outcomes, highlighting areas of elevated loss exposure. The findings support targeted strategies such as improved risk monitoring, enhanced recovery processes, and increased adoption of loan protection products to reduce future losses and strengthen portfolio performance.

Business Problem

The credit union faces recurring loan charge-offs linked to deceased members, contributing to financial losses and reduced portfolio efficiency. Despite this, there is limited analytical insight into the underlying risk factors, including loan characteristics, insurance coverage, and recovery outcomes. This lack of visibility limits the organization's ability to implement proactive risk controls, improve recovery performance, and minimize future loss exposure.

Data Overview

Customers dataset (demographics, status)

Loans dataset (loan type, balance, insurance)

Charge offs dataset (loss data)

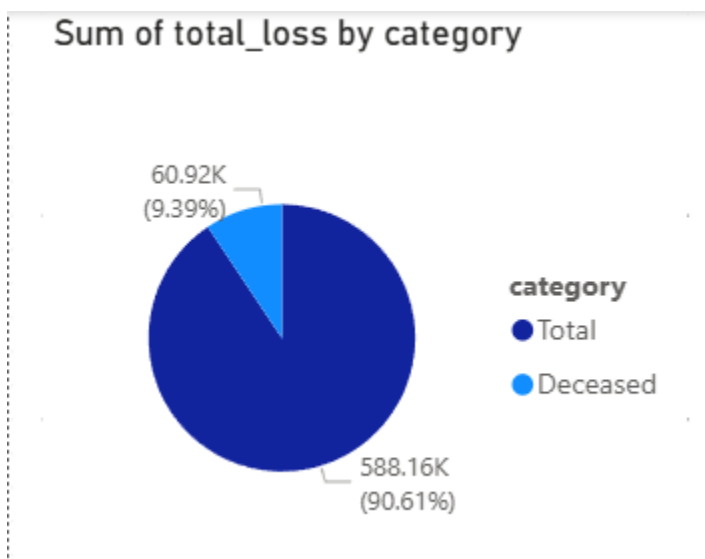
Data Cleaning

The dataset contained inconsistencies, including missing charge-off amounts and undefined loan types. These were excluded to ensure accuracy and reliability of the analysis

Key Insights

Deceased Member Contribution

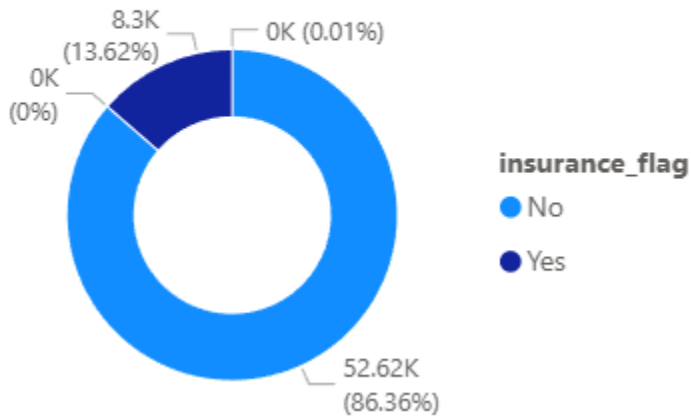
Deceased members account for approximately 10% of total charge-off losses, representing a measurable but not dominant risk segment.



Insurance coverage impact

Approximately 86% of losses are associated with uninsured loans. Highlighting a significant gap in risk protection

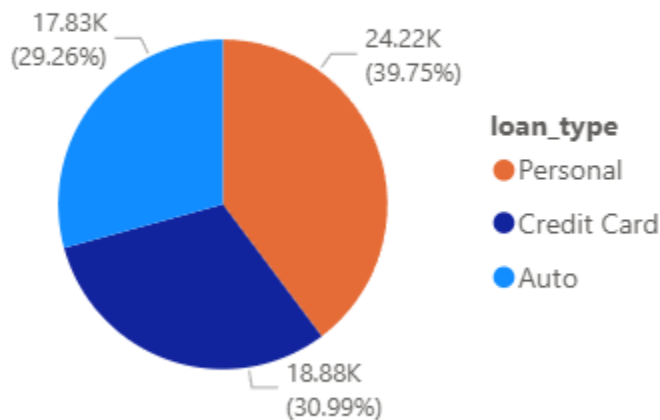
Sum of total_loans and total_loss by insurance_flag



Loan type risk

Personal loans contribute the highest losses, indicating elevated exposure to unsecured lending products.

Charge_off_amount by loan_type



Recommendations

Promote Loan Protection Insurance

- Increase adoption of credit life insurance, especially for higher-risk segments, to reduce loss exposure.

Strengthen Risk Controls for Personal Loans

- Implement stricter underwriting or insurance requirements for unsecured loans.

Improve Monitoring of High-Risk Accounts

Flag loans with:

- No insurance
- High balances
- Higher-risk profiles

Conclusion

This analysis highlights key drivers of loan charge-off and provides actionable insights to reduce financial and improve performance.